

Publication 1

Title	Predicting 30-Day Hospital Readmission in Patients with Diabetes Using Machine Learning on Electronic Health Record Data
Harvard reference	Emi-Johnson, O.G. and Nkrumah, K.J. (2025) 'Predicting 30-Day Hospital Readmission in Patients With Diabetes Using Machine Learning on Electronic Health Record Data', <i>Cureus</i> , 17(4), e82437. Available at: https://doi.org/10.7759/cureus.82437 (Accessed: 26 September 2026).
Link	https://doi.org/10.7759/cureus.82437
Modelling	Logistic Regression (LR) Random Forest (RF) XGBoost Deep Neural Network (DNN).
Preprocessing techniques	Missing numerical values were imputed using imputation. Categorical variables were one-hot encoded, while continuous variables were standardised. Records with missing or ambiguous readmission labels were removed. Class weighting was used to address class imbalance, and the data were divided into 80% training and 20% testing sets using stratified sampling.
Datasets used	Diabetes 130-US Hospitals dataset (1999–2008) dataset
Key takeaways	The study found that machine-learning models can be used to predict 30-day hospital readmission among patients with diabetes. XGBoost achieved the highest AUC-ROC (0.667) among the models tested. The study also showed the importance of addressing class imbalance and using SHAP to identify influential predictors.

Publication 2

Title	Comparison of machine-learning and logistic regression models for prediction of 30-day unplanned readmission in electronic health records: A development and validation study
Harvard reference	Iwagami, M. et al. (2024) 'Comparison of machine-learning and logistic regression models for prediction of 30-day unplanned readmission in electronic health records: A development and validation study', <i>PLOS Digital Health</i> , 3(8), e0000584. Available at: https://doi.org/10.1371/journal.pdig.0000578 (Accessed: 26 September 2026).
Link	https://doi.org/10.1371/journal.pdig.0000578
Modelling	Gradient Boosting Decision Tree (GBDT) Random Forest (RF) Deep Neural Network (DNN) Logistic Regression with LASSO (LR-LASSO).
Preprocessing techniques	Continuous variables were standardised, while categorical variables were converted into binary (dummy) variables. Variables with fewer than 10 patients were excluded, and cases with missing data were removed.
Datasets used	Electronic health records from 38 hospitals in Japan, with 339,513 discharges used for model development (2015–2017) and 118,074 discharges used for validation (2018).
Key takeaways	The study found that machine-learning models can be used to predict 30-day unplanned hospital readmission using EHR data. GBDT generally performed better than logistic regression, although the difference became smaller when more predictor variables and blood-test results were included.

Publication 3

Title	The 30-days hospital readmission risk in diabetic patients: predictive modeling with machine learning classifiers
Harvard reference	Shang, Y. et al. (2021) 'The 30-days hospital readmission risk in diabetic patients: predictive modeling with machine learning classifiers', <i>BMC Medical Informatics and Decision Making</i> , 21(Suppl 2), 57. Available at: https://doi.org/10.1186/s12911-021-01423-y (Accessed: 26 September 2026).
Link	https://doi.org/10.1186/s12911-021-01423-y
Modelling	Random Forest (RF) Naive Bayes (NB) Tree Ensemble (TE)
Preprocessing techniques	Missing values were processed, attributes with large amounts of missing data were removed, and irrelevant attributes were deleted. Missing race values were replaced using mode substitution. Records with missing diagnosis values were removed. Down-sampling, over-sampling and SMOTE were used to address class imbalance. Feature selection was also performed to reduce the number of variables.
Datasets used	Health Facts Database (Cerner Corporation, US), containing clinical records of diabetic patients from 130 hospitals in the United States from 1999 to 2008. After preprocessing, 100,244 records were included in the model construction.
Key takeaways	The study found that machine-learning models can be used to predict 30-day hospital readmission among diabetic patients. Random Forest achieved the highest AUC among the three models tested, and important predictors included number of inpatient admissions, age, diagnosis, emergency visits, and sex.

Publicly available dataset table

Title	Diabetes 130-US Hospitals for Years 1999-2008
Harvard reference	Clore, J., Cios, K., DeShazo, J. and Strack, B. (2014) <i>Diabetes 130-US Hospitals for Years 1999-2008</i> [Dataset]. UCI Machine Learning Repository. Available at: https://doi.org/10.24432/C5230J (Accessed: 26 September 2026).
Link	https://archive.ics.uci.edu/dataset/296/diabetes+130+us+hospitals+for+years+1999-2008
Number of instances	101,766 instances. No predefined train/test split.
Number of feature columns	47 feature columns
Number of classes (if applicable)	Number of classes: 3 classes • <30 = readmitted within 30 days • >30 = readmitted after 30 days • No = no recorded readmission
Target range (if applicable)	Not applicable (classification dataset).
Origin	The data comes from the Health Facts database (Cerner Corporation, USA), containing clinical records from 130 hospitals and integrated delivery networks in the United States, covering the period 1999–2008.